

Topic:-

Collect information about how silver foil (Varkha) used in sweets is made and which metals are used to make it.

Introduction:-

Whenever we visit a sweet shop during festivals like Diwali, Eid, or weddings, we see delicious Indian sweets like Kaju Katli, Besan Ladoo, and Gulab Jamun decorated with a shiny, paper-thin silver leaf. This edible metallic leaf is traditionally called 'Varkha' (वरख) or Silver Foil. It gives royal sweets an attractive decorative appearance and has been used in India for hundreds of years.

In Chapter 7 of our Class 8 Science textbook, we study the physical properties of metals. One of the most remarkable physical properties of metals is **malleability**—the property by which metals can be hammered, beaten, or rolled into ultra-thin sheets without breaking or cracking. Gold and silver are the most malleable metals in the world. In this project, we collected information on how silver foil is made, the metals used, and safety issues regarding adulteration.

❖ Notable Physical Properties of Silver (Ag):-

- **Malleability:** Pure silver can be beaten into sheets less than 0.2 microns thin!
- **Lustre:** Brilliant metallic silver-white shine that reflects light beautifully.
- **Bactericidal Nature:** Pure silver has natural anti-microbial properties that prevent sweets from spoiling quickly.

Aim & Information Sources:-

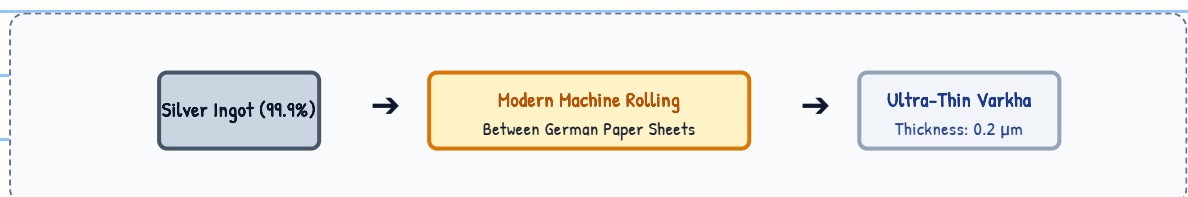
Aim: To collect authentic information regarding the traditional and modern manufacture of silver foil (varkha), analyze which metals are used, and learn how to test for adulteration.

❖ Objectives of the Project:-

- To understand the metallic property of malleability through silver varkha.
- To compare traditional manual hammering vs modern hygienic machine rolling.
- To perform simple lab tests to detect dangerous aluminum adulteration in sweets.

❖ Information Sources Used:-

- Class 8 General Science textbook (Chapter 7: Metals and Nonmetals).
- Interview with a traditional sweet maker (Halwai) in our city market.
- Food Safety and Standards Authority of India (FSSAI) official safety regulations.



How Silver Foil (Varkha) Is Made:-

Our research revealed the fascinating industrial process of creating silver foil:

- **1. Selecting Pure Silver:** High-grade silver metal of 99.9% purity (chandi) is melted in a graphite crucible and cast into thin silver strips.
- **2. Interleaving in Special Paper:** Small pieces of silver strips are placed between hundreds of sheets of specially treated vegetable parchment paper (or synthetic German sheets).
- **3. Traditional Manual Beating:** In older times, artisans called 'Varkh-saz' beat the leather pouches continuously with heavy wooden mallets for up to 8 hours until the metal flattened out into microscopic thickness.
- **4. Modern Automated Machinery:** FSSAI now mandates modern computerized rolling machines. Clean stainless steel rollers beat silver between sterile synthetic sheets without any human touch or animal parts.
- **5. Final Packaging:** The paper-thin silver leaf (about 0.15 to 0.2 microns thick) is gently transferred between booklet sheets of tissue paper.



Metals Used & Adulteration Table:-

The table below compares pure silver foil with illegal adulterants like aluminum:

Parameter	Pure Silver Varkha (Safe)	Adulterated Aluminum Foil (Toxic!)
Metal Used	99.9% Pure Silver (Ag)	Industrial Aluminum (Al) scrap
Cost	Expensive (~₹75,000 per kg)	Very cheap (~₹200 per kg)
Touch & Rub Test	Crumbles into microscopic powder immediately	Turns into thick grey metallic flakes on fingers
Flame Test	Burns instantly into tiny silvery ball; no black ash	Burns with bright spark leaving blackish ash
Chemical Acid Test	Dissolves in nitric acid (HNO ₃) forming clear liquid	Does not dissolve in HNO ₃ easily

Scientific Analysis & Health Impact:-

Our study uncovered the chemistry and health consequences of metals in food:

- **Extreme Malleability of Silver:** One gram of pure silver can be beaten into nearly 1 square meter of ultra-thin foil! Silver atoms slide smoothly over one another in metallic bonding without fracturing.
- **Biological Safety of Pure Silver:** Pure silver is chemically unreactive in the human stomach. It passes out harmlessly through the digestive tract without being absorbed into blood vessels.
- **Toxicity of Aluminum Adulteration:** Unscrupulous vendors sometimes substitute toxic aluminum foil to save money. When humans ingest aluminum, it accumulates over years in the liver, bones, and brain, causing stomach ulcers, anemia, and neurological disorders.

❖ *FSSAI Modern Regulations:-*

The Food Safety and Standards Authority of India (FSSAI) has strictly banned animal intestines in varkha production and mandates a minimum of 99.9% pure silver certification.

Everyday Applications & Quick Home Tests:-

Here are simple home tests anyone can perform to check sweet purity:

- **1. The Gentle Touch Test:** Gently touch the foil on top of the sweet with your finger. If it sticks to your finger in thick grey layers, it is fake aluminum! Pure silver leaf crumbles instantly and adheres only to the sweet.
- **2. The Matchstick Flame Test:** Scrape a tiny piece of foil onto a spoon and hold it over a candle flame. Pure silver turns into a tiny bright silver bead; aluminum leaves black residue and white smoke.
- **3. Rubbing Between Palms:** Rub the foil between both dry palms. Pure silver disappears completely into microscopic dust, while aluminum rolls into a noticeable ball.

❖ Other Uses of Silver Foil in Traditional Medicine:-

In Ayurvedic medicine, purified silver (*Rajat Bhasma*) has been used as a brain tonic and cooling agent for centuries.

What We Learned & Conclusion:-

- Silver and gold are the most malleable metals, allowing them to be hammered into microscopically thin foils.
- Authentic varkha must contain at least 99.9% pure silver (Ag) and be manufactured by modern hygienic machines.
- Adulterating sweets with cheap aluminum foil is illegal and harmful to human digestive health.
- Simple household tests (rub test and flame test) enable consumers to detect fake aluminum adulteration.

Conclusion:-

Investigating silver foil (varkha) gave us a wonderful real-life demonstration of metallic malleability. We learned how ancient traditional artistry has been modernized with hygienic food technology. Testing sweets for purity taught us that science is an essential shield protecting consumer safety and public health during festive seasons.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 7: Metals and Nonmetals).
- Food Safety and Standards Authority of India (FSSAI) Notification on Silver Leaf Regulations.
- Consumer Guidance Society of India (CGSI) Food Adulteration Testing Handbook.